

考試規範：

1. 本試卷共四大題(共一頁)，配分請見各題，合計 100 分。
2. 禁止帶手機、計算機及電腦等電子用品進入考場。
3. 所有題目的解答都要有計算過程，無計算過程不給分。

1. Let D be the plane region enclosed by the graph of curves $y = \sqrt{9 - x^2}$, $x = -1$, $y = 0$ and $y = x$. (設 D 是平面上由這些曲線的圖形圍繞而成的區域)

(1) (8 pts) Find the area of D .

(2) (6 pts) Find the volume of the solid generated by revolving D about x -axis. (求 D 繞 x 軸旋轉所生成之物體體積)

2. (6 pts) Find the area of the surface generated by revolving the curve $y = \sqrt{2x - x^2}$, $1 \leq x \leq 1.5$, about x -axis. (求曲線繞 x 軸旋轉所生成之曲面面積)

3. (8 pts×8) Evaluate the integrals. (計算積分)

(1) $\int_0^{\pi/2} \cos^3(3\theta) \sin^2(6\theta) d\theta$

(2) $\int x^3 \sin(2x) dx$

(3) $\int \frac{2x^2 - 6x + 4}{(x^2 + 1)(x - 1)^3} dx$

(4) $\int \tan^{-1}(\sqrt{x}) dx$

(5) $\int \sqrt[3]{\tan x} dx$

(6) $\int \ln(x) \cdot x^{\ln(x)-1} dx$

(7) $\int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025 - x}} dx$

(8) $\int \frac{\sin^2(x)}{1 + \cos^2(x)} dx$

4. (8 pts×2) Find the limit if it exists. (求極限若存在)

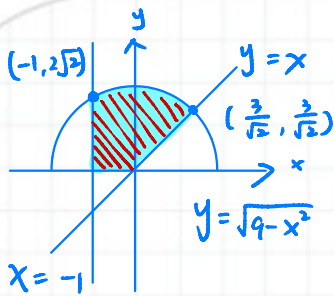
(1) $\lim_{t \rightarrow (-2)^+} \int_t^2 \frac{5t + 10}{x^2 - x - 6} dx$

(2) $\lim_{x \rightarrow 0} \left(\csc^2(x) - \frac{1}{x^2} \right)$

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11401 微積分乙期末考評分標準

題號 1 - (1)



(2分)

$$\begin{aligned} \text{Area} &= \int_{-1}^0 \sqrt{9-x^2} dx + \int_0^{\frac{3}{\sqrt{2}}} \sqrt{9-x^2} - x dx \\ &= \int_{-1}^{\frac{3}{\sqrt{2}}} \sqrt{9-x^2} dx - \int_0^{\frac{3}{\sqrt{2}}} x dx \\ &= \left. \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) + \frac{1}{2} x \sqrt{9-x^2} \right|_{-1}^{\frac{3}{\sqrt{2}}} - \left. \frac{1}{2} x^2 \right|_0^{\frac{3}{\sqrt{2}}} \\ &= \frac{9}{2} \cdot \frac{\pi}{4} + \frac{1}{2} \cdot \frac{3}{\sqrt{2}} \cdot \frac{3}{\sqrt{2}} - \frac{9}{2} \sin^{-1}\left(\frac{1}{3}\right) - \frac{1}{2} (-1) \cdot 2\sqrt{2} - \frac{9}{4} \\ &= \frac{9}{8} \pi - \frac{9}{2} \sin^{-1}\left(\frac{1}{3}\right) + \sqrt{2} \end{aligned}$$

(2分)

or $\frac{9}{2} \sin^{-1}\left(\frac{1}{3}\right)$ or $\frac{9}{2} \cos^{-1}\left(\frac{\sqrt{8}}{3}\right)$ or $\frac{9}{2} \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

Area = $\int_{-1}^0 \sqrt{9-x^2} dx + \frac{3}{2} \pi \times \frac{1}{8}$

列式正確 2分 $\rightarrow$ $= \int_{\sin^{-1}(\frac{1}{3})}^0 9 \cdot \left(\frac{1+\cos 2\theta}{2}\right) d\theta + \frac{9}{8} \pi$

代換正確 2分 $\rightarrow$ $= \frac{9}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_{\sin^{-1}(\frac{1}{3})}^0 + \frac{9}{8} \pi$

上下界代換 正確 2分 $\rightarrow$ $= \frac{9}{2} \left(0 + 0 - \sin^{-1}\left(\frac{1}{3}\right) - \frac{1}{2} \cdot 2 \cdot \left(\frac{1}{3}\right) \cdot \frac{2\sqrt{2}}{3} \right) + \frac{9}{8} \pi$

答案正確 2分 $\rightarrow$ $= -\frac{9}{2} \sin^{-1}\left(\frac{1}{3}\right) + \sqrt{2} + \frac{9}{8} \pi$

$$\begin{aligned} \text{Area} &= \int_{-1}^{\frac{3}{\sqrt{2}}} \sqrt{9-x^2} dx - \frac{3}{2} \cdot \frac{3}{\sqrt{2}} \times \frac{1}{2} \\ &= \frac{9}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_{\sin^{-1}(\frac{1}{3})}^{\frac{\pi}{4}} - \frac{9}{4} \\ &= \frac{9}{2} \left[\frac{\pi}{4} + \frac{1}{2} \cdot 2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \sin^{-1}\left(\frac{1}{3}\right) - \frac{1}{2} \cdot 2 \cdot \left(\frac{1}{3}\right) \cdot \left(\frac{2\sqrt{2}}{3}\right) \right] - \frac{9}{4} \\ &= \frac{9}{8} \pi - \frac{9}{2} \sin^{-1}\left(\frac{1}{3}\right) + \sqrt{2} \end{aligned}$$

$$\int \sqrt{9-x^2} dx \quad 4分$$

Let $\sin \theta = \frac{x}{3} \Rightarrow x = 3 \sin \theta$
 $dx = 3 \cos \theta d\theta$

$$= \int 3 \cos \theta \cdot 3 \cos \theta d\theta$$

$$= 9 \int \cos^2 \theta d\theta$$

$$= 9 \int \frac{1+\cos 2\theta}{2} d\theta$$

$$= \frac{9}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right] + C$$

$$= \frac{9}{2} \left[\sin^{-1}\left(\frac{x}{3}\right) + \frac{1}{2} \cdot 2 \cdot \sin \theta \cdot \cos \theta \right] + C$$

$$= \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) + \frac{9}{2} \cdot \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3} + C$$

$$= \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) + \frac{1}{2} x \sqrt{9-x^2} + C$$

- 只有圖正確 1分，
圖正確 + 列式正確 2分
- 使用三角代換且正確 2分，
換回 x 正確 2分。
- 上下界代入，答案正確 2分。

$\frac{2\sqrt{2}}{1} + \frac{1}{\sin^{-1}(\frac{1}{3})} + 45^\circ$

$$= \frac{2\sqrt{2} \times 1}{2} + \frac{1}{2} \cdot 3 \cdot \sin^{-1}\left(\frac{1}{3}\right) + \frac{1}{2} \cdot 3 \cdot \frac{\pi}{4}$$

$$= \sqrt{2} + \frac{9}{2} \sin^{-1}\left(\frac{1}{3}\right) + \frac{9\pi}{8}$$

$\Delta + \text{扇形} - \text{扇形} \rightarrow \cos^{-1}\left(\frac{1}{3}\right)$

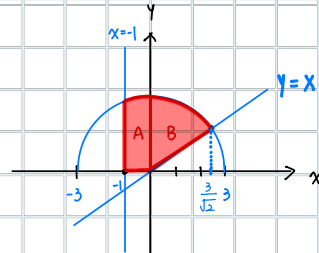
$$\sqrt{2} + \frac{27}{8} \pi - \frac{9}{2} \cos^{-1}\left(\frac{1}{3}\right)$$

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題號 1 (2)

法一：A+B



寫出來 +1分，
答案算對 +1分

寫出來 +1分，
答案算對 +1分

$$V = \int_{-1}^0 \pi (\sqrt{9-x^2})^2 dx + \int_0^{\frac{3}{\sqrt{2}}} \pi (\sqrt{9-x^2})^2 - \pi x^2 dx \quad \text{整個寫出來 +3分}$$

$$= \int_{-1}^0 \pi (\sqrt{9-x^2})^2 dx + \int_0^{\frac{3}{\sqrt{2}}} \pi (\sqrt{9-x^2})^2 dx - \int_0^{\frac{3}{\sqrt{2}}} \pi x^2 dx$$

$$= \int_{-1}^{\frac{3}{\sqrt{2}}} \pi (\sqrt{9-x^2})^2 dx - \int_0^{\frac{3}{\sqrt{2}}} \pi x^2 dx$$

$$= \pi \int_{-1}^{\frac{3}{\sqrt{2}}} (9-x^2) dx - \pi \int_0^{\frac{3}{\sqrt{2}}} x^2 dx$$

$$= \pi \left(9x - \frac{1}{3}x^3 \right) \Big|_{-1}^{\frac{3}{\sqrt{2}}} - \pi \left(\frac{1}{3}x^3 \right) \Big|_0^{\frac{3}{\sqrt{2}}} \quad +1 \text{分 (積分正確)}$$

$$= \pi \left[\left(\frac{27}{\sqrt{2}} - \frac{1}{3} \frac{27}{2\sqrt{2}} \right) - \left(-9 + \frac{1}{3} \right) \right] - \pi \left(\frac{1}{3} \frac{27}{2\sqrt{2}} - 0 \right) \quad +1 \text{分 (數字代入且都正確)}$$

$$= \pi \left[\frac{27}{\sqrt{2}} - \frac{9}{2\sqrt{2}} + 8\frac{2}{3} - \frac{9}{2\sqrt{2}} \right]$$

$$= \pi \left(\frac{27}{\sqrt{2}} - \frac{9}{\sqrt{2}} + 8\frac{2}{3} \right)$$

$$= \pi \left(\frac{18}{\sqrt{2}} + 8\frac{2}{3} \right)$$

$$= \pi \left(9\sqrt{2} + 8\frac{2}{3} \right) \quad \# \quad 1 \text{分}$$

正常給分為紅色部分，

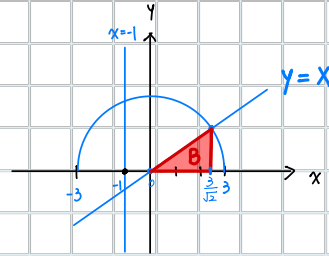
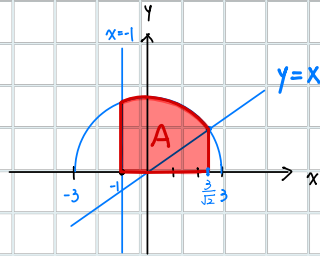
若寫出橘色部分會部分給分，但公式及積分範圍要寫對！

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題號 1 (2)

法二：A - B



寫出來 + 1 分，
答案算對 + 1 分

寫出來 + 1 分，
答案算對 + 1 分

$$V = \int_{-1}^{\frac{\sqrt{2}}{2}} \pi (\sqrt{9-x^2})^2 dx - \int_0^{\frac{\sqrt{2}}{2}} \pi x^2 dx \quad \text{整個寫出來 + 3 分}$$

$$= \pi \int_{-1}^{\frac{\sqrt{2}}{2}} (9-x^2) dx - \pi \int_0^{\frac{\sqrt{2}}{2}} x^2 dx$$

$$= \pi \left(9x - \frac{1}{3}x^3 \right) \Big|_{-1}^{\frac{\sqrt{2}}{2}} - \pi \left(\frac{1}{3}x^3 \right) \Big|_0^{\frac{\sqrt{2}}{2}} \quad + 1 \text{ 分 (積分正確)}$$

$$= \pi \left[\left(\frac{27}{\sqrt{2}} - \frac{1}{3} \frac{27}{2\sqrt{2}} \right) - \left(-9 + \frac{1}{3} \right) \right] - \pi \left(\frac{1}{3} \frac{27}{2\sqrt{2}} - 0 \right) \quad + 1 \text{ 分 (數字代入且都正確)}$$

$$= \pi \left[\frac{27}{\sqrt{2}} - \frac{9}{2\sqrt{2}} + 8\frac{2}{3} - \frac{9}{2\sqrt{2}} \right]$$

$$= \pi \left(\frac{27}{\sqrt{2}} - \frac{9}{\sqrt{2}} + 8\frac{2}{3} \right)$$

$$= \pi \left(\frac{18}{\sqrt{2}} + 8\frac{2}{3} \right)$$

$$= \pi \left(9\sqrt{2} + 8\frac{2}{3} \right) \quad + 1 \text{ 分}$$

正常給分為紅色部分，

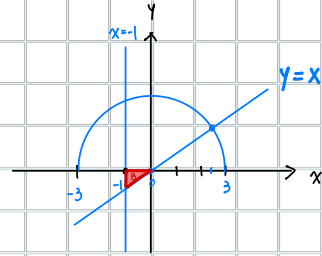
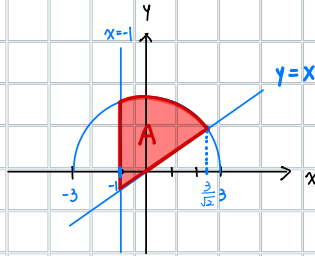
若寫出橘色部分會部分給分，但公式及積分範圍要寫對！

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題號 1 (2)

法三：A - B



寫出來 +1分，
答案算對 +1分

寫出來 +1分，
答案算對 +1分

$$V = \int_{-1}^{\frac{3}{\sqrt{2}}} \pi (\sqrt{9-x^2})^2 - \pi x^2 dx - \int_{-1}^0 (0^2 - x^2) \pi dx \quad \text{整個寫出來 +3分}$$

$$= \pi \int_{-1}^{\frac{3}{\sqrt{2}}} (9 - 2x^2) dx - \pi \int_{-1}^0 -x^2 dx$$

$$= \pi \left(9x - \frac{2}{3}x^3 \right) \Big|_{-1}^{\frac{3}{\sqrt{2}}} + \pi \left(\frac{1}{3}x^3 \right) \Big|_{-1}^0 \quad +1分 (積分正確)$$

$$= \pi \left[\left(\frac{27}{\sqrt{2}} - \frac{2}{3} \frac{27}{\sqrt{2}} \right) - \left(-9 + \frac{2}{3} \right) \right] + \pi \left[0 - \left(-\frac{1}{3} \right) \right] \quad +1分 (數字代入且都正確)$$

$$= \pi \left[\frac{27}{\sqrt{2}} - \frac{9}{\sqrt{2}} + 8\frac{1}{3} + \frac{1}{3} \right]$$

$$= \pi \left(\frac{18}{\sqrt{2}} + 8\frac{2}{3} \right)$$

$$= \pi (9\sqrt{2} + 8\frac{2}{3}) \quad +1分$$

正常給分為紅色部分，

若寫出橘色部分會部分給分，但公式及積分範圍要寫對！

請用粗水性筆填寫

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題號 2

$$y = \sqrt{2x - x^2} \Rightarrow y' = \frac{dy}{dx} = \frac{1}{2\sqrt{2x - x^2}} \cdot (2 - 2x) \\ = \frac{(1-x)}{\sqrt{2x - x^2}} \quad (1)$$

$$A = \int_1^{1.5} 2\pi y \sqrt{1 + (y')^2} dx \\ = \int_1^{1.5} 2\pi \sqrt{2x - x^2} \cdot \sqrt{1 + \frac{(1-x)^2}{2x - x^2}} dx \quad (2) \\ = \int_1^{1.5} 2\pi \sqrt{2x - x^2 + (1 - 2x + x^2)} dx \\ = \int_1^{1.5} 2\pi dx = 2\pi x \Big|_{x=1}^{1.5} = \pi \quad (1)$$

114-1 微積分乙期末考第 3 題第 1 小題 Question 3(1)

3.(1) Evaluate the integral:

$$\int_0^{\frac{\pi}{2}} \cos^3(3\theta) \sin^2(6\theta) d\theta$$

Solution:

$$\begin{aligned} \text{Let } I &= \int_0^{\frac{\pi}{2}} \cos^3(3\theta) \cdot [2 \sin(3\theta) \cos(3\theta)]^2 d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} \cos^5(3\theta) \sin^2(3\theta) d\theta \end{aligned}$$

Method 1: Apply substitution: Let $u = \sin(3\theta)$, then $du = 3 \cos(3\theta) d\theta \implies \cos(3\theta) d\theta = \frac{du}{3}$.
Consider limits: When $\theta = 0, u = 0$; when $\theta = \frac{\pi}{2}, u = \sin(\frac{3\pi}{2}) = -1$.
Rewriting $\cos^4(3\theta) = (1 - \sin^2(3\theta))^2 = (1 - u^2)^2$:

$$\begin{aligned} I &= 4 \int_0^{-1} (1 - u^2)^2 \cdot u^2 \cdot \frac{du}{3} = \frac{4}{3} \int_0^{-1} (1 - 2u^2 + u^4) u^2 du = \frac{4}{3} \int_0^{-1} (u^2 - 2u^4 + u^6) du = \frac{4}{3} \left[\frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} \right]_0^{-1} \\ &= \frac{4}{3} \left(\frac{-1}{3} - \frac{2(-1)}{5} + \frac{-1}{7} \right) = \frac{4}{3} \left(\frac{-1}{3} + \frac{2}{5} - \frac{1}{7} \right) = \frac{4}{3} \left(\frac{-35 + 42 - 15}{105} \right) = \frac{4}{3} \left(\frac{-8}{105} \right) = -\frac{32}{315} \end{aligned}$$

Method 2: Rewrite integrand using $\sin^2(3\theta) = 1 - \cos^2(3\theta)$.

$$I = 4 \int_0^{\frac{\pi}{2}} \cos^5(3\theta) (1 - \cos^2(3\theta)) d\theta = 4 \int_0^{\frac{\pi}{2}} [\cos^4(3\theta) - \cos^6(3\theta)] \cos(3\theta) d\theta$$

Apply substitution: Let $u = \sin(3\theta)$, then $du = 3 \cos(3\theta) d\theta \implies \cos(3\theta) d\theta = \frac{du}{3}$.
Consider limits: When $\theta = 0, u = 0$; when $\theta = \frac{\pi}{2}, u = \sin(\frac{3\pi}{2}) = -1$.
Rewriting terms: $\cos^4(3\theta) = (1 - u^2)^2$ and $\cos^6(3\theta) = (1 - u^2)^3$.

$$\begin{aligned} I &= 4 \int_0^{-1} [(1 - u^2)^2 - (1 - u^2)^3] \cdot \frac{du}{3} \\ &= \frac{4}{3} \int_0^{-1} (1 - u^2)^2 \cdot [1 - (1 - u^2)] du \\ &= \frac{4}{3} \int_0^{-1} (1 - u^2)^2 \cdot u^2 du = \frac{4}{3} \int_0^{-1} (1 - 2u^2 + u^4) u^2 du \\ &= \frac{4}{3} \int_0^{-1} (u^2 - 2u^4 + u^6) du = \frac{4}{3} \left[\frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} \right]_0^{-1} \\ &= \frac{4}{3} \left(\frac{-1}{3} - \frac{2(-1)}{5} + \frac{-1}{7} \right) = \frac{4}{3} \left(\frac{-8}{105} \right) = -\frac{32}{315} \end{aligned}$$

Table 1: 評分標準

作答情況	得分	詳細說明
正確解法	8	過程完整。正確完成三角整理、變數變換 (u -substitution)、上下限轉換正確 ($0 \rightarrow -1$)，且最終答案 $\frac{-32}{315}$ 正確。
正確過程：計算失誤	6	觀念完全正確 (包含積分上下限轉換正確)，僅在最終多項式積分或分數加減運算中出現數值錯誤。
Wallis 公式	Max 6	使用 Wallis 公式求解且答案正確。因本題旨在測驗變數變換與積分技巧，且該公式非課本核心範圍，故給予過程分 4 分，答案分 2 分，上限 6 分。
觀念錯誤：邊界未換	Max 4	三角函數整理正確 (2 分)，且變數變換設定正確 ($u = \sin 3\theta$)，但忽略積分上下限轉換 (如維持 $0 \rightarrow \frac{\pi}{2}$)。導致後續邏輯錯誤，後續分數不予計算。
僅三角函數整理	2	僅正確將原式化簡為 $4 \int \cos^5(3\theta) \sin^2(3\theta) d\theta$ ，後續未作答或完全錯誤。
僅引用公式	1	僅正確寫出二倍角公式 $\sin(6\theta) = 2 \sin(3\theta) \cos(3\theta)$ ，無有效後續推導。
完全錯誤/空白	0	空白或算式與題目無關。

Table 2: Grading Rubric

Scenario	Score	Description & Criteria
Standard Solution: Perfect	8	Complete and correct process. Successfully performed trigonometric simplification, correct u -substitution, correct change of limits ($0 \rightarrow -1$), and obtained the correct final answer ($\frac{-32}{315}$).
Standard Solution: Calculation Error	6	Concepts are fully correct (including the correct change of integration limits). The error is strictly limited to arithmetic or final polynomial integration.
Non-Standard Method: Wallis' Formula	Max 6	Used Wallis' formula to obtain the correct answer. Since the exam assesses substitution and integration techniques, and this formula is outside the core curriculum, the score is capped (4 pts for process, 2 pts for answer).
Conceptual Error: Limits Not Changed	Max 4	Correct trigonometric simplification (2 pts) and valid setup for substitution ($u = \sin 3\theta$), but failed to update the limits of integration (e.g., kept $0 \rightarrow \frac{\pi}{2}$). Subsequent logic is considered invalid.
Trigonometric Simplification Only	2	Only correctly simplified the integrand to $4 \int \cos^5(3\theta) \sin^2(3\theta) d\theta$, with no further valid progress.
Formula Citation Only	1	Only correctly stated the double-angle formula $\sin(6\theta) = 2 \sin(3\theta) \cos(3\theta)$, without effective application.
Incorrect / Blank	0	Blank answer or irrelevant calculations.

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題號 3(2)

Q 3(2). $\int x^3 \sin 2x dx$

D	I
x^3	$\sin 2x$
$3x^2$	$-\frac{1}{2} \cos 2x$
$6x$	$-\frac{1}{4} \sin 2x$
6	$\frac{1}{8} \cos 2x$
0	$\frac{1}{16} \sin 2x$

① 大註 = 忘了寫 C 扣 1 分

② 忘了轉回去 x 扣 2 分

$$A = \underbrace{-\frac{1}{2} x^3 \cos 2x}_{+2} + \underbrace{\frac{3}{4} x^2 \sin 2x}_{+2} + \underbrace{\frac{3}{4} x \cos 2x}_{+2} - \underbrace{\frac{3}{8} \sin 2x}_{+2} + C$$

(II) Let $u = 2x$, $du = 2dx \Rightarrow \frac{1}{2} \int \left(\frac{u}{2}\right)^3 \sin u du$

$$= \left(\frac{1}{2}\right)^4 \int u^3 \sin u du$$

D	I
u^3	$\sin u$
$3u^2$	$-\cos u$
$6u$	$-\sin u$
6	$\cos u$
0	$\sin u$

$$\text{所以} = \frac{1}{16} [-u^3 \cos u + 3u^2 \sin u + 6u \cos u - 6 \sin u + C]$$

$$= \underbrace{-\frac{1}{2} x^3 \cos 2x}_{+2} + \underbrace{\frac{3}{4} x^2 \sin 2x}_{+2} + \underbrace{\frac{3}{4} x \cos 2x}_{+2} - \underbrace{\frac{3}{8} \sin 2x}_{+2} + C$$

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題號 3 (3)

$$3 (3) \quad \frac{2x^2-6x+4}{(x^2+1)(x-1)^3} = \frac{2(x-1)(x-2)}{(x^2+1)(x-1)^{\cancel{3}^2}} = \frac{2x-4}{(x^2+1)(x-1)^2}$$

$$\text{設 } \frac{2x-4}{(x^2+1)(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+1}$$

$$\text{或 } \frac{2x^2-6x+4}{(x^2+1)(x-1)^3} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{E}{(x-1)^3} + \frac{Cx+D}{x^2+1}$$

2分

經解聯立後得知

$$A=2, B=-1, E=0, C=-2, D=-1$$

4分

(p.s.) 無解聯立過程 -2 ; 係數錯一個 -1

$$\text{所求} = 2 \int \frac{1}{x-1} dx - \int \frac{1}{(x-1)^2} dx + \int \frac{-2x-1}{x^2+1} dx$$

$$= 2 \ln|x-1| + (x-1)^{-1} - \ln|x^2+1| - \tan^{-1}x + C, C \in \mathbb{R}$$

最後一部分共2分，沒有寫常數 -1

其餘部分較特別處為：如果前面解聯

立時係數已算錯，但後續積分計算沒問題時，還是有最後這一部分的分數

請用粗水性筆填寫

11401 微積分乙期末考評分標準

題號 3(4)

Metho1 $\int \arctan(\sqrt{x}) dx$ Use integration by parts.

$$= \underline{x \arctan(\sqrt{x})} - \int x \cdot \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} dx$$

2 points 2 points

$$= x \arctan(\sqrt{x}) - \int \frac{x}{1+x} \cdot \frac{dx}{2\sqrt{x}}$$

• To calculate $\int \frac{x}{1+x} \cdot \frac{dx}{2\sqrt{x}}$

Let $y = \sqrt{x}$, $dy = \frac{1}{2\sqrt{x}} dx$,

2 points

$$\begin{aligned} \text{Then } \int \frac{x}{1+x} \cdot \frac{dx}{2\sqrt{x}} &= \int \frac{y^2}{1+y^2} dy \\ &= \int \left(1 - \frac{1}{1+y^2}\right) dy \\ &= y - \arctan y + C \\ &= \underline{\sqrt{x} - \arctan(\sqrt{x})} + C \end{aligned}$$

1 point

$$\begin{aligned} \text{Hence, } x \arctan(\sqrt{x}) - \int \frac{x}{1+x} \cdot \frac{dx}{2\sqrt{x}} \\ = \underline{x \arctan(\sqrt{x}) - \sqrt{x} + \arctan(\sqrt{x})} + C \end{aligned}$$

1 point

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題號 3(4)

(Method 2) Let $\sqrt{x} = \tan \theta \Rightarrow \frac{dx}{2\sqrt{x}} = \sec^2 \theta d\theta$
 $\Rightarrow dx = 2 \tan \theta \sec^2 \theta d\theta$
2 points.

Then $\int \arctan(\sqrt{x}) dx = \int 2\theta \tan \theta \sec^2 \theta d\theta$

$= \frac{\theta \tan^2 \theta}{2 \text{ points}} - \frac{\tan \theta + \theta}{2 \text{ points}} + C$

Integration by part table

D	I
2θ	$\tan \theta \sec^2 \theta$
2	$\frac{1}{2} \tan^2 \theta$
0	$\frac{1}{2} \tan \theta - \frac{1}{2} \theta$

$= \frac{x \arctan(\sqrt{x}) - \sqrt{x} + \arctan(\sqrt{x})}{2 \text{ points}} + C$

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題號 3(5) 法一

Let $u = \tan^{\frac{1}{3}} x \Rightarrow u^3 = \tan x$ 變數變換 2分

$\Rightarrow x = \tan^{-1}(u^3) \Rightarrow dx = \frac{3u^2}{1+u^6} du$

1分

$\Rightarrow \int \tan^{\frac{1}{3}} x dx = \int \frac{3u^3}{1+u^6} du$

Let $v = u^2$

$\Rightarrow dv = 2u du$

$v = \tan^{\frac{2}{3}} x$

$= \frac{1}{2} \int \frac{3v}{1+v^3} dv$

1分

$= \frac{1}{2} \int \frac{3v}{(v+1)(v^2-v+1)} dv = \frac{1}{2} \int \frac{-1}{v+1} + \frac{v+1}{v^2-v+1} dv$

$= \frac{1}{2} \int \frac{-1}{v+1} + \frac{\frac{1}{2}(2v-1) + \frac{3}{2}}{v^2-v+1} dv$

2分

$= \frac{1}{2} \int \frac{-1}{v+1} + \frac{1}{2} \cdot \frac{2v-1}{v^2-v+1} + \frac{3}{2} \cdot \frac{1}{(v-\frac{1}{2})^2 + \frac{3}{4}} dv$

$= \frac{1}{2} \left(-\ln|v+1| + \frac{1}{2} \ln|v^2-v+1| + \frac{3}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2}{\sqrt{3}}(v-\frac{1}{2})\right) \right) + C$

$= -\frac{1}{2} \ln|\tan^{\frac{2}{3}} x + 1| + \frac{1}{4} \ln|\tan^{\frac{4}{3}} x - \tan^{\frac{2}{3}} x + 1|$ 答案 2分

$+ \frac{\sqrt{3}}{2} \tan^{-1}\left(\frac{2}{\sqrt{3}}(\tan^{\frac{2}{3}} x - \frac{1}{2})\right) + C$

#

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題號 3(5) 法二

$$\text{Let } u = \tan x \Rightarrow du = \sec^2 x dx \quad 1 \text{分}$$

$$\Rightarrow du = (1 + \tan^2 x) dx \Rightarrow dx = \frac{1}{1+u^2} du$$

$$\Rightarrow \int \tan^{\frac{1}{3}} x dx = \int \frac{u^{\frac{1}{3}}}{1+u^2} du$$

$$\frac{1}{\text{分}} \left(\int \frac{3v^2}{1+v^6} dv = \frac{1}{2} \int \frac{3w}{1+w^3} dw \right)$$

$$\begin{aligned} \text{let } v &= u^{\frac{1}{3}} \\ \Rightarrow v^3 &= u \\ \Rightarrow du &= 3v^2 dv \end{aligned} \quad 1 \text{分}$$

$$\begin{aligned} \text{let } w &= v^3 \\ \Rightarrow dw &= 3v^2 dv \\ w &= \tan^{\frac{2}{3}} x \end{aligned} \quad 1 \text{分}$$

$$= \frac{1}{2} \int \frac{3w}{(w+1)(w^2-w+1)} dw \quad 2 \text{分}$$

= 同法 - ...

$$= \frac{1}{2} \left(-\ln|w+1| + \frac{1}{2} \ln|w^2-w+1| + \frac{3}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2}{\sqrt{3}} \left(w - \frac{1}{2} \right) \right) \right) + C$$

$$= -\frac{1}{2} \ln|\tan^{\frac{2}{3}} x + 1| + \frac{1}{4} \ln|\tan^{\frac{4}{3}} x - \tan^{\frac{2}{3}} x + 1| \quad \text{答案} \quad 2 \text{分}$$

$$+ \frac{\sqrt{3}}{2} \tan^{-1} \left(\frac{2}{\sqrt{3}} (\tan^{\frac{2}{3}} x - \frac{1}{2}) \right) + C$$

#

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題號 3(6)

$$\int \ln x \cdot x^{\ln x - 1} dx$$

$$= \int x^{\ln x} \cdot \frac{\ln x}{x} dx$$

2分

$$= \int (e^u)^u \cdot u du = \int u \cdot e^{u^2} du$$

$$= \frac{1}{2} \int e^v dv$$

答案2分

$$= \frac{1}{2} e^v + C = \frac{1}{2} e^{(\ln x)^2} + C$$

$$= \frac{1}{2} x^{\ln x} + C$$

#

$$\text{let } u = \ln x$$

$$du = \frac{1}{x} dx$$

$$x = e^u$$

2分

$$\text{let } v = u^2$$

$$dv = 2u du$$

2分

$$v = (\ln x)^2$$

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題號 3(7)-1

Sol 1 (標準做法)

Let $u = 2025 - x \Rightarrow du = -dx$, $2025 - 2025 = 0$, $2025 - 0 = 2025$ ②

$$\begin{aligned} \int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}} dx &= \int_{2025}^0 \frac{\sqrt{2025-u}}{\sqrt{2025-u} + \sqrt{u}} (-1) du = \int_0^{2025} \frac{\sqrt{2025-u}}{\sqrt{2025-u} + \sqrt{u}} du \\ &= \int_0^{2025} \frac{\sqrt{2025-u} + \sqrt{u} - \sqrt{u}}{\sqrt{2025-u} + \sqrt{u}} du = \int_0^{2025} 1 - \frac{\sqrt{u}}{\sqrt{2025-u} + \sqrt{u}} du \end{aligned}$$

$$\text{Let } u = x, du = dx, \text{ and } I = \int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}} dx = \int_0^{2025} \frac{\sqrt{u}}{\sqrt{u} + \sqrt{2025-u}} du$$

$$\text{Then, } I = \int_0^{2025} 1 du - I \Rightarrow \underline{2I = \int_0^{2025} 1 du} \quad ④$$

$$\Rightarrow I = \frac{1}{2} \int_0^{2025} 1 du = \frac{1}{2} [x]_0^{2025} = \underline{\frac{2025}{2}} \quad \# \quad ③$$

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題號 3(7)-2

Sol 2 (偏方: 積分定義)

Let $f(x) = \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}}$, $\int_0^{2025} f(x) dx$ is the area for $x \in [0, 2025]$

Observe $f(x)$ when $x \in [0, 2025]$

If $x=0$, $f(0) = 0$

If $x=2025$, $f(2025) = 1$

If $x = \frac{0+2025}{2}$, $f(\frac{2025}{2}) = \frac{\sqrt{\frac{2025}{2}}}{\sqrt{\frac{2025}{2}} + \sqrt{\frac{2025}{2}}} = \frac{1}{2} \rightarrow$ middle point

maybe it is symmetric?

And for $x_1 = \frac{2025}{2} + c$, $x_2 = \frac{2025}{2} - c$, $\forall c \in [0, \frac{2025}{2}]$

$$\begin{aligned} f(x_1) + f(x_2) &= \frac{\sqrt{\frac{2025}{2} + c}}{\sqrt{\frac{2025}{2} + c} + \sqrt{\frac{2025}{2} - c}} + \frac{\sqrt{\frac{2025}{2} - c}}{\sqrt{\frac{2025}{2} - c} + \sqrt{\frac{2025}{2} + c}} \\ &= \frac{\frac{2025}{2} + c - \sqrt{(\frac{2025}{2})^2 - c^2}}{\frac{2025}{2} + c - \frac{2025}{2} + c} + \frac{\sqrt{(\frac{2025}{2})^2 - c^2} - \frac{2025}{2} + c}{\frac{2025}{2} + c - \frac{2025}{2} + c} \\ &= \frac{2c}{2c} = 1 \end{aligned}$$

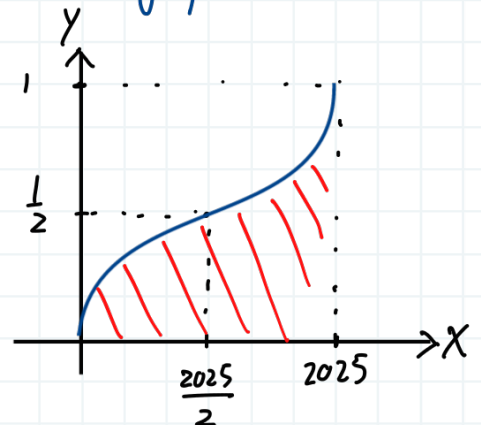
So it's area can be written as

high of $f(\frac{2025}{2} + c) + f(\frac{2025}{2} - c)$

$$\int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}} dx = \frac{2025}{2} \cdot 1 = \frac{2025}{2}$$

↑
range of c

i.e. The graph is like



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題號 3(7)-3

Sol 3 (有理化 + 三角代換)

$$\int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}} dx = \int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}} \cdot \frac{\sqrt{x} - \sqrt{2025-x}}{\sqrt{x} - \sqrt{2025-x}} dx = \int_0^{2025} \frac{x - \sqrt{2025x-x^2}}{2x - 2025} dx \quad \textcircled{1}$$

$$= \int_0^{2025} \frac{x - \sqrt{\left(\frac{2025}{2}\right)^2 - \left(\frac{2025}{2}\right)^2 + 2025x - x^2}}{2x - 2025} dx = \int_0^{2025} \frac{x - \sqrt{\left(\frac{2025}{2}\right)^2 - \left(x - \frac{2025}{2}\right)^2}}{2x - 2025} dx$$

$$\text{let } x - \frac{2025}{2} = \frac{2025}{2} \sin \theta, \quad dx = \frac{2025}{2} \cos \theta d\theta,$$

$$\frac{2025}{2} = \frac{2025}{2} \sin \theta \Rightarrow \theta = \frac{\pi}{2}, \quad 0 - \frac{2025}{2} = \frac{2025}{2} \sin \theta = -\frac{\pi}{2}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\frac{2025}{2} \sin \theta + \frac{2025}{2} - \frac{2025}{2} \sqrt{1 - \sin^2 \theta}}{\frac{4050}{2} \sin \theta + \frac{4050}{2} - \frac{4050}{2}} \cdot \frac{2025}{2} \cos \theta d\theta$$

$$= \frac{2025}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin \theta + 1 - \cos \theta}{\sin \theta} \cdot \cos \theta d\theta = \frac{2025}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta + \frac{(1 - \cos \theta) \cos \theta}{\sin \theta} d\theta \quad \textcircled{2}$$

Consider θ as $-\theta$, $\frac{(1 - \cos(-\theta)) \cos(-\theta)}{\sin(-\theta)} = \frac{(1 - \cos \theta) \cos \theta}{-\sin \theta} \Rightarrow$ it is odd function

$$= \frac{2025}{4} \left([\sin \theta] \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + 0 \right) = \frac{2025}{4} \cdot 2 = \frac{2025}{2} \quad \textcircled{2}$$

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題號 3(7)-4

Sol 4 (三角代換 + 標準做法) ← 也繞一大圈

Let $x = 2025 \sin^2 \theta$, $dx = 4050 \sin \theta \cos \theta d\theta$, $2025 = 2025 \sin^2 \theta \Rightarrow \theta = \frac{\pi}{2}$, $0 = 2025 \sin^2 \theta \Rightarrow \theta = 0$

$$\begin{aligned} \int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}} dx &= \int_0^{\frac{\pi}{2}} \frac{45 \sin \theta}{45 \sin \theta + 45 \cos \theta} \cdot 4050 \sin \theta \cos \theta d\theta = 4050 \int_0^{\frac{\pi}{2}} \frac{\sin^2 \theta \cos \theta}{\sin \theta + \cos \theta} d\theta \\ &= 4050 \int_0^{\frac{\pi}{2}} \frac{\cos \theta - \cos^3 \theta}{\sin \theta + \cos \theta} d\theta = 4050 \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} - \frac{\cos^3 \theta}{\sin \theta + \cos \theta} d\theta \end{aligned} \quad (2)$$

Substitute θ as $\frac{\pi}{2} - \theta$, $\cos(\frac{\pi}{2} - \theta) = \sin \theta$

$$\begin{aligned} \text{Then, } I &= \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\cos \theta + \sin \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{\sin \theta}{\cos \theta + \sin \theta} d\theta \\ \Rightarrow I + I &= \int_0^{\frac{\pi}{2}} \frac{\cos \theta + \sin \theta}{\cos \theta + \sin \theta} d\theta \Rightarrow \underline{I = \frac{\pi}{4}} \quad (1) \end{aligned}$$

$$\text{And, } I = \int_0^{\frac{\pi}{2}} \frac{\cos^3 \theta}{\cos \theta + \sin \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{\sin^3 \theta}{\cos \theta + \sin \theta} d\theta$$

$$\Rightarrow I + I = \int_0^{\frac{\pi}{2}} \frac{\cos^3 \theta + \sin^3 \theta}{\cos \theta + \sin \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{(\cos^2 \theta + \sin^2 \theta - \sin \theta \cos \theta)(\cos \theta + \sin \theta)}{\cos \theta + \sin \theta} d\theta$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 - \sin \theta \cos \theta d\theta = \frac{1}{2} \left[\theta - \frac{1}{4} \cos 2\theta \right]_0^{\frac{\pi}{2}} = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 - \frac{1}{2} \sin(2\theta) d\theta$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - \frac{1}{4} - \frac{1}{4} \right) = \underline{\frac{\pi}{4} - \frac{1}{4}} \quad (1)$$

$$\begin{aligned} \text{Therefore, } \int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025-x}} dx &= 4050 \left(\int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\cos \theta + \sin \theta} d\theta - \int_0^{\frac{\pi}{2}} \frac{\cos^3 \theta}{\cos \theta + \sin \theta} d\theta \right) \\ &= 4050 \left(\frac{\pi}{4} - \frac{\pi}{4} + \frac{1}{4} \right) = \underline{\frac{2025}{2}} \quad (2) \end{aligned}$$

114-1 期末考評分標準 題號3(8)

法一：

$$\begin{aligned}\int \frac{\sin^2 x}{1 + \cos^2 x} dx &= \int \frac{1 - \cos^2 x}{1 + \cos^2 x} dx = \int \frac{2 - (1 + \cos^2 x)}{1 + \cos^2 x} dx \\ &= \int \frac{2}{1 + \cos^2 x} - 1 dx = 2 \int \frac{1}{\sin^2 x + 2 \cos^2 x} dx - x + C_1 \\ &= 2 \int \frac{\frac{1}{\cos^2 x}}{\frac{\sin^2 x}{\cos^2 x} + 2} dx - x + C_1 = 2 \int \frac{\sec^2 x}{\tan^2 x + 2} dx - x + C_1\end{aligned}$$

$$\text{Let } u = \tan x, du = \sec^2 x dx$$

$$= 2 \int \frac{du}{u^2 + 2} - x + C_1 = \int \frac{du}{\left(\frac{u}{\sqrt{2}}\right)^2 + 1} - x + C_1$$

$$\text{Let } v = \frac{u}{\sqrt{2}}, dv = \frac{du}{\sqrt{2}}$$

$$= \sqrt{2} \int \frac{dv}{v^2 + 1} - x + C_1 = \sqrt{2} \tan^{-1} v + C_2 - x + C_1$$

$$= \sqrt{2} \tan^{-1} \left(\frac{u}{\sqrt{2}} \right) - x + C = \sqrt{2} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) - x + C$$

評分標準：

1. 開頭有使用任何三角函數相關公式換過給1分，最多給2分。
2. 積分使用代換，並且代換正確給2分，最多給4分。
3. 最終答案錯誤，但有給出 $-x + C$ 或 $\int -1 dx$ 等形式，給1分，答案完全正確給2分。

法二：

$$\int \frac{\sin^2 x}{1 + \cos^2 x} dx = \int \frac{\tan^2 x}{\sec^2 x + 1} dx$$

$$\text{Let } u = \tan x, du = \sec^2 x dx = (1 + u^2) dx$$

$$= \int \frac{u^2}{u^2 + 2} \frac{1}{u^2 + 1} du = \int \frac{2}{u^2 + 2} + \frac{-1}{u^2 + 1} du$$

$$= \sqrt{2} \tan^{-1} \left(\frac{u}{\sqrt{2}} \right) - \tan^{-1} u + C$$

$$= \sqrt{2} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) - \tan^{-1}(\tan x) + C$$

$$= \sqrt{2} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) - x + C$$

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題號

4.(1) 法1

$$\text{原式} = \lim_{t \rightarrow (-2)^+} \left[\overset{\text{提出來}}{(5t+10)} \cdot \int_t^2 \frac{1}{x^2-x-6} dx \right]$$

$$= \lim_{t \rightarrow (-2)^+} \left[\frac{\int_t^2 \frac{1}{x^2-x-6} dx}{\left(\frac{1}{5t+10}\right)} \right] \quad \text{不定型 } \frac{\infty}{\infty}$$

+1分

+1分

$$\stackrel{\text{L.H.}}{=} \lim_{t \rightarrow (-2)^+} \frac{\cancel{(-1)} \cdot \frac{1}{t^2-t-6}}{\frac{+5}{(5t+10)^2}}$$

後拉到分母均不給分

+1分

$$= \lim_{t \rightarrow (-2)^+} \frac{5 \cdot 25(t+2)^2}{5(t+2)(t-3)}$$

$$= \frac{5 \cdot (-2+2)}{-2-3} = \frac{0}{-5} = 0$$

+1分

★1

$$\lim_{t \rightarrow -2^+} \int_t^2 \frac{1}{x^2 - x - 6} dx = \lim_{t \rightarrow -2^+} \int_t^2 \frac{1}{(x-3)(x+2)} dx$$

$$= \lim_{t \rightarrow -2^+} \int_t^2 \frac{\frac{1}{5}[(x+2) - (x-3)]}{(x-3)(x+2)} dx$$

$$= \lim_{t \rightarrow -2^+} \frac{1}{5} \left[\int_t^2 \frac{1}{x-3} - \frac{1}{x+2} \right] dx$$

±2分

$$= \lim_{t \rightarrow -2^+} \frac{1}{5} \left[\ln|x-3| - \ln|x+2| \right]_t^2$$

±1分

$$= \lim_{t \rightarrow -2^+} \frac{1}{5} (\ln 1 - \ln 4 - \ln|t-3| + \ln|t+2|)$$

GR

$= -\infty$

±1分

(這個值一定要出來)

※沒說明★1但答案算對者至多4分

※題目5t+10寫成5x+10一律0分

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題號 4.(1) 法2

$$\text{原式} = \lim_{t \rightarrow (-2)^+} \int_t^2 \frac{5t+10}{(x-3)(x+2)} dx$$

$$\frac{5t+10}{(x-3)(x+2)} = \frac{A}{(x-3)} + \frac{B}{x+2}$$

$$\Rightarrow 5t+10 = A(x+2) + B(x-3)$$

$$x=-2 \text{ 代入: } 5t+10 = 0 + (-5)B$$

$$\Rightarrow B = -t-2$$

$$x=3 \text{ 代入: } 5t+10 = 5A + 0$$

$$\Rightarrow A = t+2$$

+2分

此部分佔 2 分，若將
t 視為 x 或是有計算錯誤
則此題一律 0 分!!

$$= \lim_{t \rightarrow -2^+} \int_t^2 \frac{t+2}{x-3} - \frac{(t+2)}{x+2} dx$$

$$= \lim_{t \rightarrow -2^+} (t+2) \left[\ln|x-3| - \ln|x+2| \right] \Big|_t^2$$

12分

$$= \lim_{t \rightarrow -2^+} \underbrace{(t+2)}_{\rightarrow 0} \cdot \left[\underbrace{\ln 1 - \ln 2}_{\text{GR}} - \underbrace{\ln|t-3|}_{\nearrow 0} + \underbrace{\ln|t+2|}_{\rightarrow \infty} \right]$$

$$\star | \lim_{t \rightarrow -2^+} (t+2) \ln(t+2) = \lim_{t \rightarrow -2^+} \frac{\ln(t+2)}{\frac{1}{t+2}} \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L.H.}}{=} \lim_{t \rightarrow -2^+} \frac{\frac{1}{t+2}}{\frac{-1}{(t+2)^2}}$$

$$= \lim_{t \rightarrow -2^+} -(t+2)$$

$$= 0$$

12分

$$= 0 + 0$$

$$= 0$$

12分

P.S. $\star |$ 內容必寫
若沒寫將不給此部分
分和最終 answer 答案的
分數!!

P.S.: 給分已
經很寬鬆
請不要為難
各班助教
3Q~

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題號 4.(2) 選

$$\text{原式} = \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x}{x^2 \sin^2 x} \quad \text{不定型 } \frac{0}{0}$$

+2分

$$= \lim_{x \rightarrow 0} \frac{x^2 (x + \sin x)(x - \sin x)}{x^4 \sin^2 x}$$

+1分

$$= \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x} \cdot \frac{x + \sin x}{x} \cdot \frac{x - \sin x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x} \cdot \lim_{x \rightarrow 0} \left(1 + \frac{\sin x}{x} \right) \cdot \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \quad \text{不定型 } \frac{0}{0}$$

$$= \frac{1}{1} \cdot (1+1) \cdot \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} \quad \text{L.H.} \quad \text{不定型 } \frac{0}{0}$$

1分

1分

L.H. = 2

$$= 2 \cdot \lim_{x \rightarrow 0} \frac{\sin x}{6x} \quad \text{1分}$$

1分

$$= 2 \cdot \frac{1}{6} = \frac{1}{3} \quad \text{2分}$$

2分

材料記
羅必達
扣1分.

✱若一直合法羅必達
但沒算出 answer
至多+4分

請用粗水性筆填寫

11401 微積分乙期末考評分標準

題號 4. (2) 法2

$$\text{原式} = \lim_{x \rightarrow 0} \frac{x^2 \csc^2 x - 1}{x^2} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{\sin^2 x} - 1}{x^2}$$

這項一定要出來!
+2分

不定型 $\frac{0}{0}$

寫這項後直接羅必達不給分
 $\because 0 \cdot \infty = ?$

$$\text{L.H.} = \lim_{x \rightarrow 0} \frac{\cancel{2x}(\sin^2 x) - \cancel{x} \cdot \cancel{2} \sin x \cos x}{\cancel{2x}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x - x \sin x \cos x}{\sin^3 x}$$

+2分

$$= \lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{\sin^3 x}$$

不定型 $\frac{0}{0}$

$$\text{L.H.} = \lim_{x \rightarrow 0} \frac{\cancel{\cos x} - (\cancel{\cos x} - \cancel{x} \sin x)}{3 \sin^2 x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{x}{3 \sin x \cos x}$$

+2分

$$= \lim_{x \rightarrow 0} \frac{1}{3 \cos x} \cdot \frac{x}{\sin x}$$

$$= \frac{1}{3} \cdot 1 = \frac{1}{3}$$

+2分

標記羅必達扣1分
至多扣2分

P.S.: 給分已經很寬鬆
請不要為難
各班助教
3Q~